

Multiple Choice (10 marks)

1. What proportion of vegans is likely to be deficient in vitamin B₁₂?
 - (a) 3%
 - (b) 32%
 - (c) 66%
 - (d) 15%
2. Nitrites are added to processed meats mainly to prevent the growth of the following genus of microorganisms
 - (a) salmonella
 - (b) clostridium
 - (c) listeria
 - (d) campylobacter
3. Colonic microbiome:
 - (a) Is present in babies in utero
 - (b) Is a cause of eosinophilic oesophagitis
 - (c) Is easily changed by probiotic supplementation
 - (d) May be changed with consumption of oligo-saccharides
4. Which of the following diet-derived compounds have epigenetic effects?
 - (a) Butyrate
 - (b) Curcumin
 - (c) Genistein
 - (d) All of the options given are correct
5. What nutritional strategies will not improve performance during prolonged endurance type exercise?
 - (a) Carbohydrate loading
 - (b) Tapering
 - (c) The classic glycogen loading regime
 - (d) Exogenous carbohydrate ingestion during exercise

True / False (10 marks)

Answer True or False

6. True or False: Rickets of prematurity is associated with hypocalcaemic convulsions, use of frusemide diuretic, and maternal vitamin D deficiency.
7. True or False: A deficiency of folate can lead to megaloblastic anaemia, a condition characterized by larger than normal red blood cells.
8. True or False: Randomised controlled trials are the most effective way to study the relationship between sugar intake and dental caries risk.
9. True or False: Energy balance in a healthy person is usually achieved on a day-to-day basis.
10. True or False: A woman should eat less than 200 kCal per day during pregnancy to maintain a healthy weight.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how infections or inflammatory states can affect the accuracy of biochemical indices used to assess micronutrient status, focusing on the acute phase reaction's impact.
12. Discuss the role of goitrogens in the diet, particularly focusing on specific foods that contain them, and analyze their potential effects on thyroid health.

End of Paper